Rotary Wing Aerodynamics (AE 667)

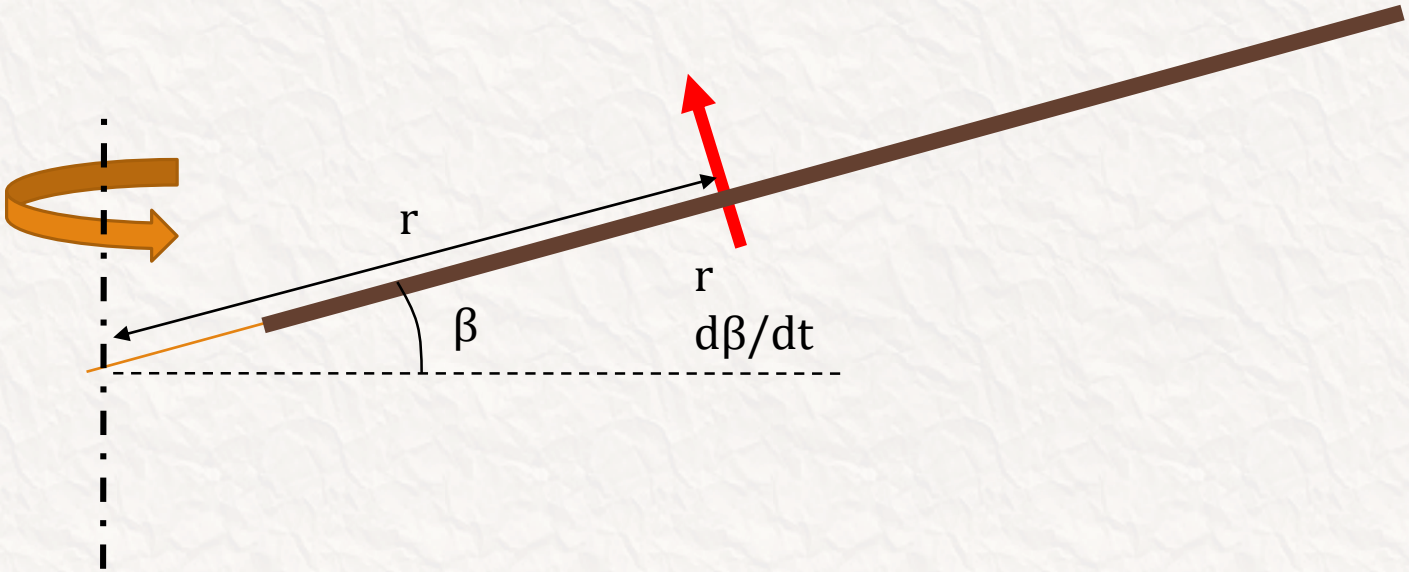


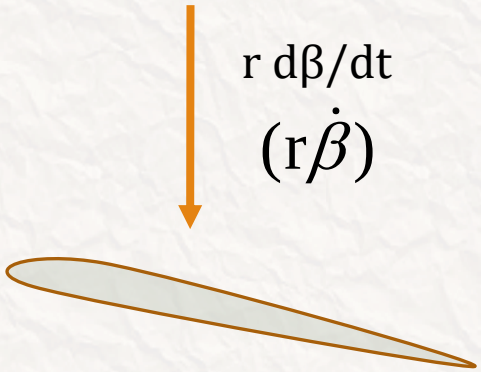
Angle of Attack & Sectional Load Calculations

Factors affecting Angle of Attack of an Airfoil

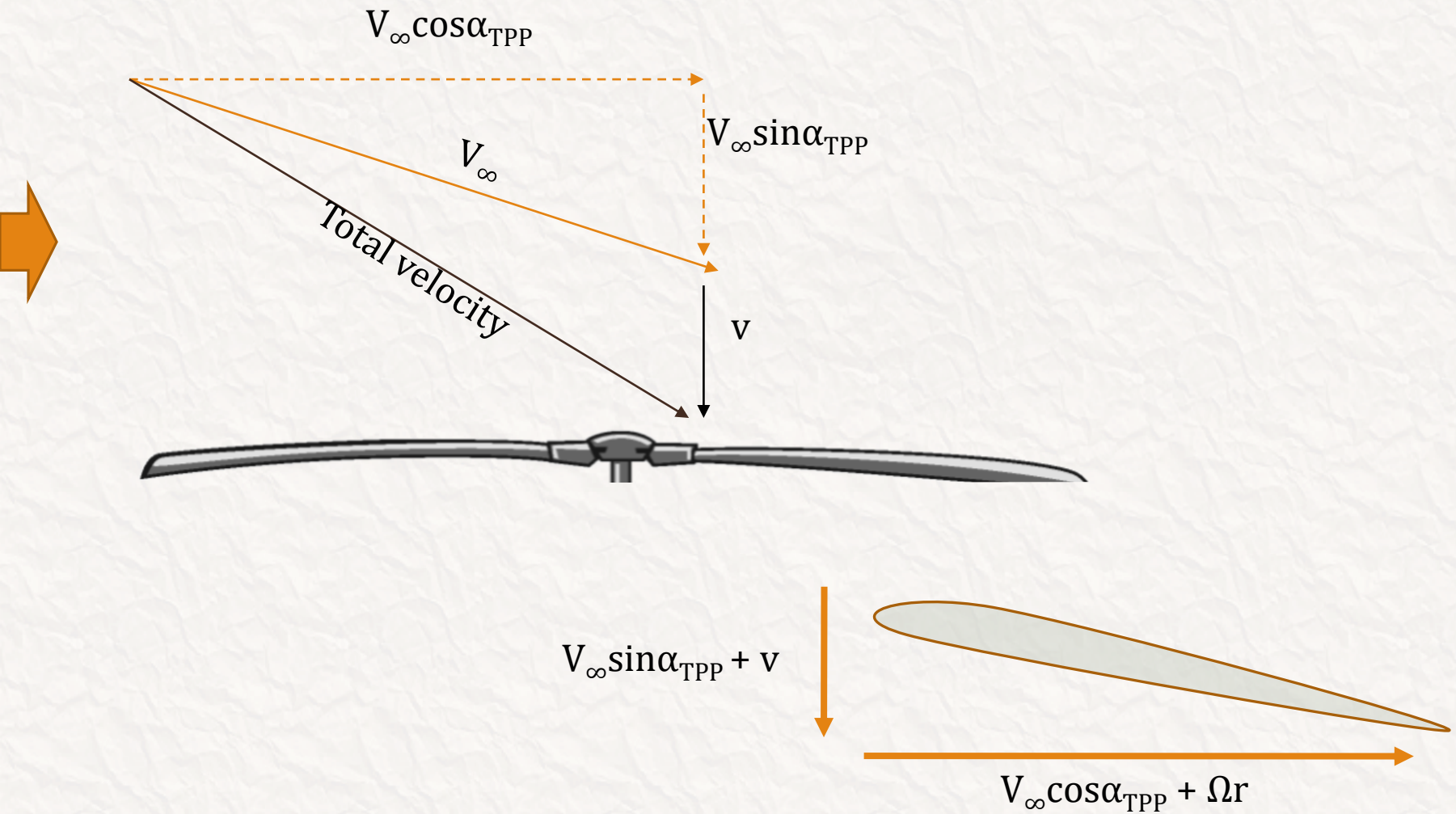
- Pilot Input: Collective & Cyclic Pitch
- Blade Twist
- Inflow: Free stream + induced velocity
- Due to blade flapping
- Anhedral/Dihedral effects due to coning

Blade flapping and Angle of Attack

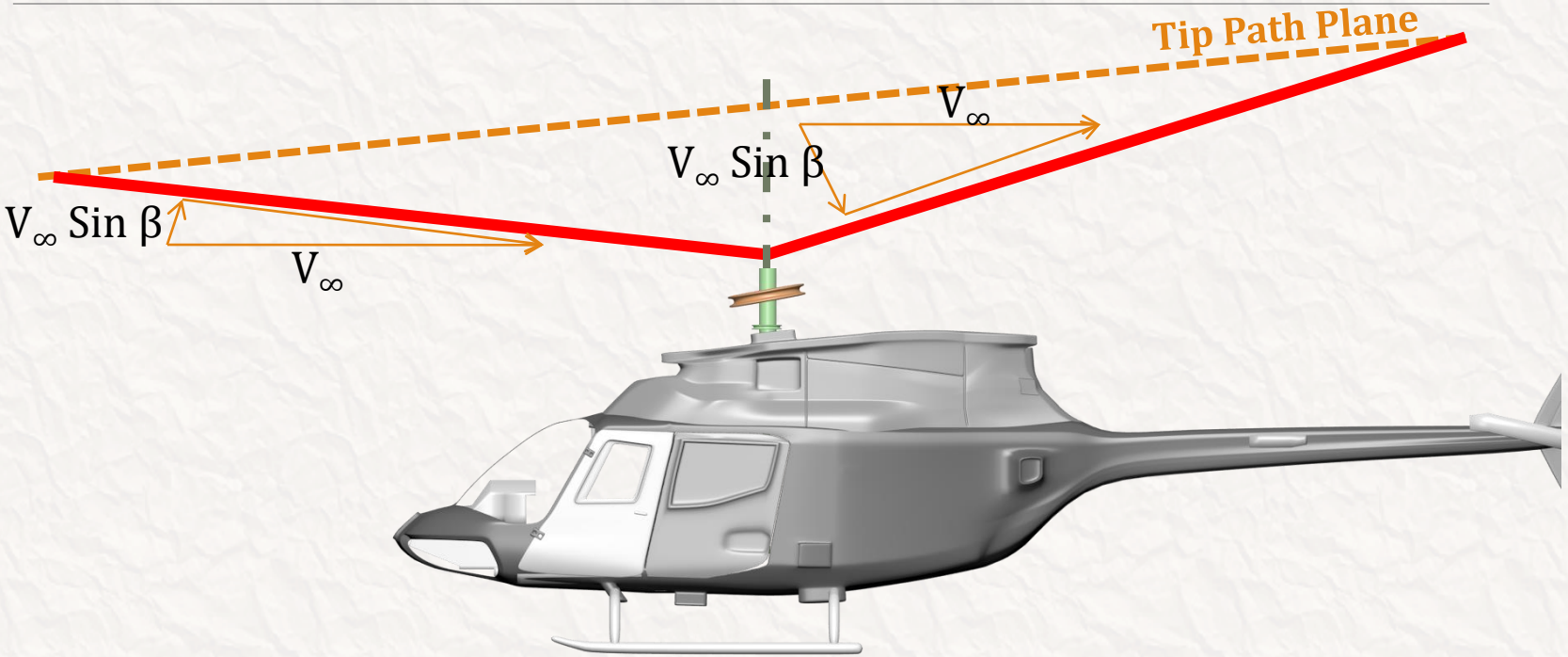




Free Steam & Induced Velocities and Angle of Attack



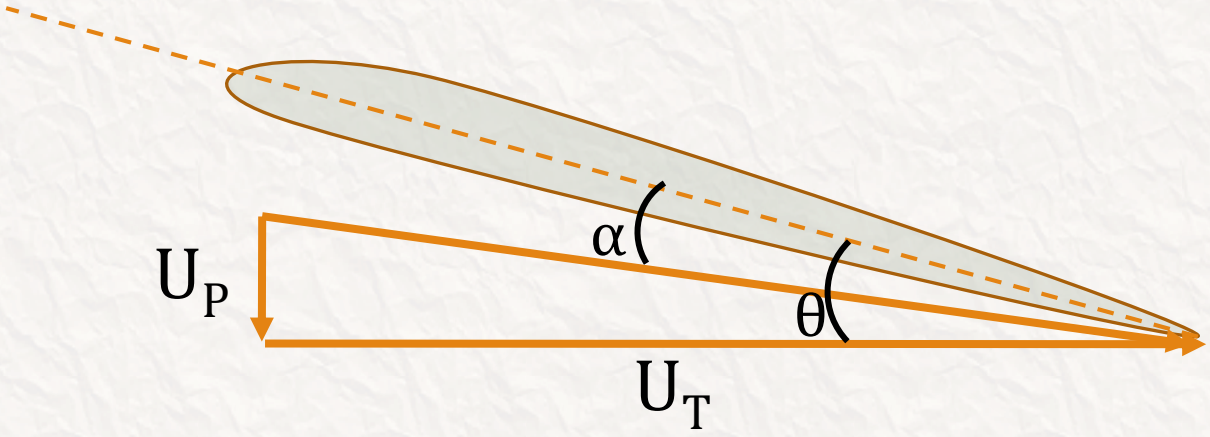
Anhedral/Dihedral Effect on AoA



Upwash at Ψ = 180° equals to V_∞ Sin β and
Downwash at Ψ = 0° equal to V_∞ Sin β.

Generalizing for all azimuths, **Downwash = V_∞ Sin β Cos Ψ**

Combining all velocities and finding AoA



- $U_P = V_\infty \text{ Sin } \alpha_{TPP} + v + r\, d\beta/dt + V_\infty \text{ Sin } \beta \text{ Cos } \Psi$
- $U_T = \Omega r + V_\infty \cos\alpha_{TPP} \text{ Sin } \Psi$

- Angle of Attack (α) = $\theta - \tan^{-1} [U_P/U_T]$

  (here, α_{TPP} refers to Tip Path Plane tilt w.r.t free stream)

Small angle assumptions

- $\mathrm{Cos}(\alpha_{TPP}) = \mathrm{Cos}(\beta) = \mathrm{Cos}(\theta) \approx 1;$
- $\mathrm{Sin}(\alpha_{TPP}) \approx \alpha_{TPP};\ \mathrm{Sin}(\beta) \approx \beta;\ \mathrm{Sin}(\theta) \approx \theta$
- $\alpha = \theta - \tan^{-1}[U_P/U_T] \approx \theta - [U_P/U_T] = (U_T\,\theta - U_P)/U_T]$

  (where $\theta = \theta_0 + \theta_{tw}(r/R) + \theta_{1c}\,\mathrm{Cos}\,\psi + \theta_{1s}\,\mathrm{Sin}\,\psi$)

- $U_T = \Omega r + V_\infty \cos\alpha_{TPP}\,\mathrm{Sin}\,\Psi \approx \Omega r + V_\infty \mathrm{Sin}\,\Psi$
- $U_P = V_\infty\,\mathrm{Sin}\,\alpha_{TPP} + v + r\,d\beta/dt + V_\infty\,\mathrm{Sin}\,\beta\,\mathrm{Cos}\,\Psi$

  $= \Omega R\lambda + r\beta' + V_\infty\,\mathrm{Sin}\,\beta\,\mathrm{Cos}\,\Psi \qquad \approx \Omega R\lambda_s + r\beta' + V_\infty\,\beta\,\mathrm{Cos}\,\Psi$

  where $\lambda = (V_\infty\,\mathrm{Sin}\,\alpha_{TPP} + v)/\Omega R$

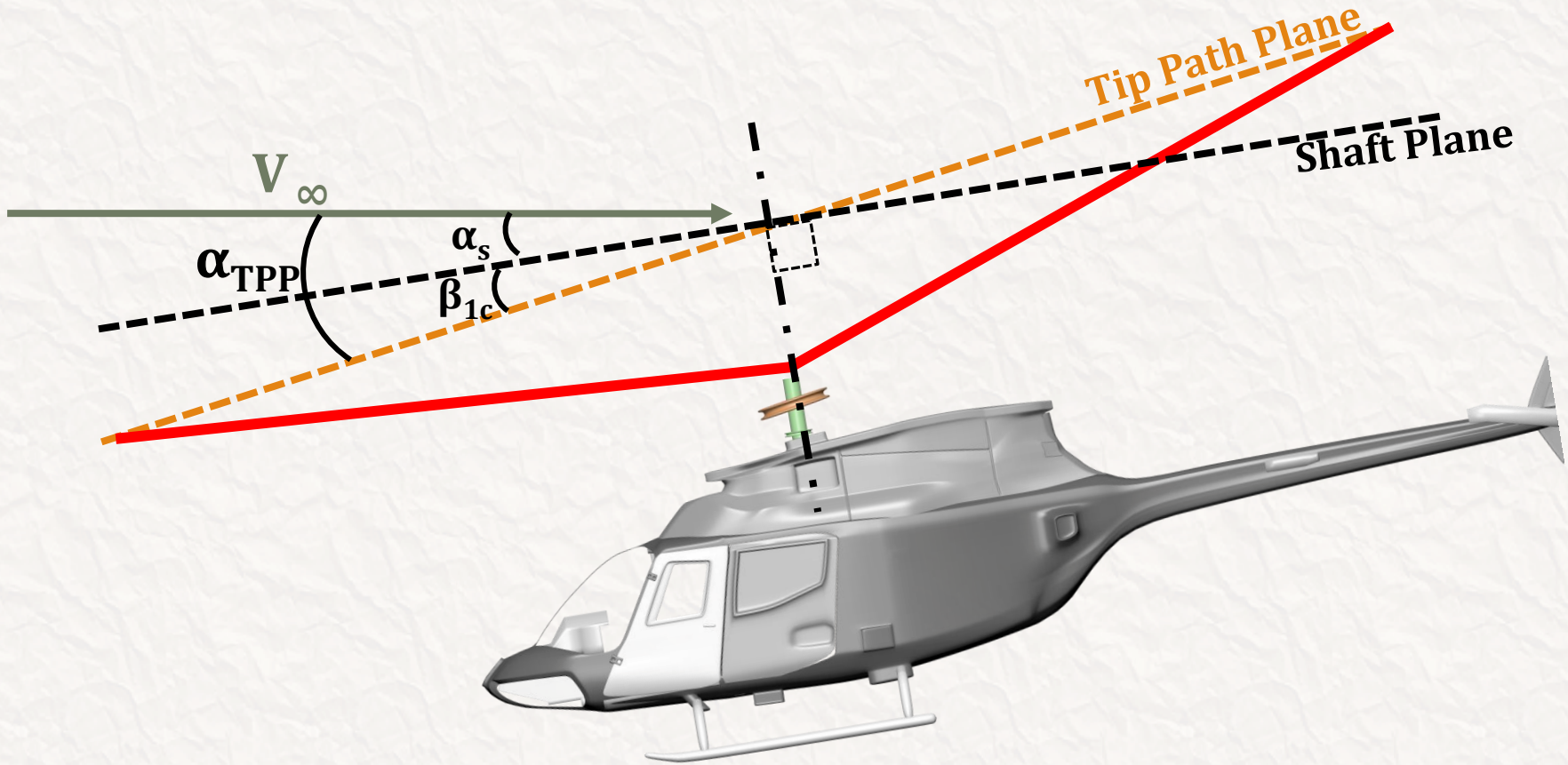
  and $\lambda_s = (V_\infty\,\mathrm{Sin}\,\alpha_{TPP})/\Omega R$

Doing some math

$$\alpha_{effective} = \frac{U_T\theta - U_P}{U_T} = \frac{1}{U_T}\left[\begin{array}{l} \Omega r\left\{\theta_0 + \theta_{tw}\frac{r}{R} + (\theta_{1c} - \beta_{1s})\cos\psi + (\theta_{1s} + \beta_{1c})\sin\psi\right\} + \\ V\left(\theta_0 + \theta_{tw}\frac{r}{R}\right)\sin\psi + V(\theta_{1c} - \beta_{1s})\cos\psi\sin\psi \\ +V(\theta_{1s} + \beta_{1c})\sin^2\psi - V\beta_0\cos\psi \\ -V\alpha_{TPP} - \mathrm{v} \end{array}\right]$$

Once α is computed at each section, finding lift, drag, pitching moment, etc. at the section can be done through airfoil tables (or empirical fits). These can be integrated over blade span to find overall thrust, torque, pitching and rolling moments.

α_{TPP} and α_s



$$\alpha_{TPP} = \alpha_s + \beta_{1c}$$

Getting Total Loads

Motive and Approach

- Need to get
  - Net thrust
  - Drag
  - In-plane side forces
  - Pitching moment
  - Rolling moment
  - Torque for a rotor

  in forward flight

- Approach: Integrating the sectional forces and moments numerically (or analytically for simplified cases)

For Analytical Integration

Assumptions:

- Constant chord (c = const.) (untapered)
- Uniform Inflow (v = const.)
- C_d = constant
- Linear Twist
- No cutout, no tip losses

Sectional Lift

$$\alpha_{effective} = \frac{U_T\theta - U_P}{U_T} = \frac{1}{U_T}\left[\begin{array}{l} \Omega r\left\{\theta_0 + \theta_{tw}\frac{r}{R} + (\theta_{1c} - \beta_{1s})\cos\psi + (\theta_{1s} + \beta_{1c})\sin\psi\right\} + \\ V\left(\theta_0 + \theta_{tw}\frac{r}{R}\right)\sin\psi + V(\theta_{1c} - \beta_{1s})\cos\psi\sin\psi \\ +V(\theta_{1s} + \beta_{1c})\sin^2\psi - V\beta_0\cos\psi \\ -V\alpha_{TPP} - \mathrm{v} \end{array}\right]$$

$$L' = \frac{1}{2}\rho c C_l\left(U_T^2 + U_P^2\right) \approx \frac{1}{2}\rho c a U_T^2\alpha_{eff} = \frac{1}{2}\rho c a\left(\theta U_T^2 - U_T U_P\right)$$

Sectional Lift

$$L' \equiv \frac{1}{2}\rho c a \left[\begin{array}{c} (\Omega r + V_\infty \sin\psi)^2 \left(\theta_0 + \theta_{tw}\frac{r}{R} + \theta_{1c}\cos\psi + \theta_{1s}\sin\psi\right) \\ -(\Omega r + V_\infty \sin\psi)(\Omega R \lambda_{TPP} + V_\infty \beta \cos\psi) \end{array} \right]$$



$$= \frac{1}{2}\rho c a \left[\begin{array}{c} \theta_0\Omega^2 r^2 + 2V_\infty \Omega r \theta_0 \sin\psi + \theta_0 V_\infty^2 \sin^2\psi + \theta_{tw}\Omega^2 \frac{r^3}{R} + 2V_\infty \Omega \theta_{tw}\frac{r^2}{R}\sin\psi + \\ \theta_{tw}\frac{r}{R}V_\infty^2 \sin^2\psi + \Omega^2 r^2 \theta_{1c}\cos\psi + 2\Omega r V_\infty \theta_{1c}\sin\psi\cos\psi + \\ V_\infty^2 \theta_{1c}\sin^2\psi\cos\psi + \\ \Omega^2 r^2 \theta_{1s}\sin\psi + 2\Omega r V_\infty \theta_{1s}\sin^2\psi + V_\infty^2 \theta_{1s}\sin^3\psi \\ -\Omega^2 R\lambda_{TPP} r - V_\infty \Omega R \lambda_{TPP}\sin\psi - \Omega r V_\infty \beta\cos\psi - V_\infty^2 \beta\cos\psi\sin\psi \end{array} \right]$$

The 2nd and 3rd harmonics in the equation translate to the vibratory loads

Total Thrust

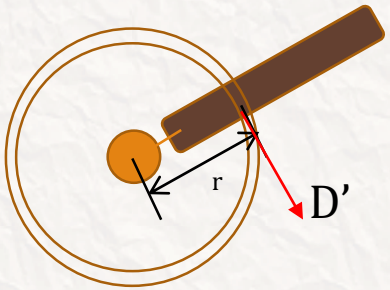
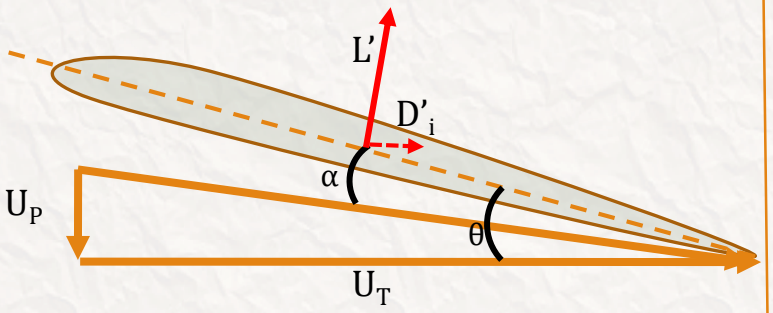
$$T = b\left[\frac{1}{2\pi}\int_0^{2\pi}\left(\int_0^R L'dr\right)d\psi\right]$$

Analytical integration using the assumptions will get us:

$$T = \frac{1}{2}\rho abc\left[\begin{array}{l}\theta_0\Omega^2\frac{R^3}{3}+\frac{\theta_0}{2}V_\infty^2 R+\theta_{tw}\Omega^2\frac{R^3}{4}+\frac{\theta_{tw}}{4}V_\infty^2 R+\\ \Omega V_\infty\theta_{1s}\big|_{TPP}\frac{R^2}{2}-\Omega^2\lambda_{TPP}\frac{R^2}{2}\end{array}\right]$$

$$C_T = \frac{\sigma a}{2}\left[\frac{\theta_0}{3}\left\{1+\frac{3}{2}\mu^2\right\}+\frac{\theta_{tw}}{4}\left\{1+\mu^2\right\}+\mu\frac{\theta_{1s}\big|_{TPP}}{2}-\frac{\lambda_{TPP}}{2}\right]$$

Torque & Power

	Profile	Induced
		
Drag:	$D' = \frac{1}{2}\rho U_T^2 c C_{d,0}$	$D'_i \cong L' \frac{U_P}{U_T} = \frac{1}{2}\rho c C_l U_T^2 \frac{U_P}{U_T}$
Torque:	$Q_0 = b\left[\frac{1}{2\pi}\int_0^{2\pi}\left\{\int_0^R rD'dr\right\}\right]$	$Q_i = \frac{b}{2\pi}\int_0^{2\pi}\left[\int_0^R rD_i dr\right]d\psi$
Power:	$P_0 = b\left[\frac{1}{2\pi}\int_0^{2\pi}\left\{\int_0^R U_T D'dr\right\}\right]$	$P_i = \frac{b}{2\pi}\int_0^{2\pi}\left[\int_0^R U_T D_i dr\right]d\psi$

Torque & Power

	Profile	Induced
C_Q	$C_{Q,0} = \frac{\sigma C_{d,0}}{8}\left[1+\mu^2\right]$	
C_P	$C_{P,0} = \frac{\sigma C_{d,0}}{8}\left[1+3\mu^2\right]$	$C_{P_i} = \lambda_i C_T$

Rotor Moments

- Overall Pitching Moment:

$$\frac{b}{2\pi}\int_0^{2\pi}\left[\int_0^R L'r\cos\psi\,dr\right]d\psi$$

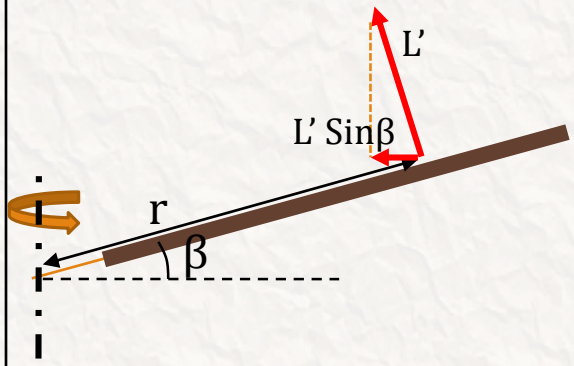
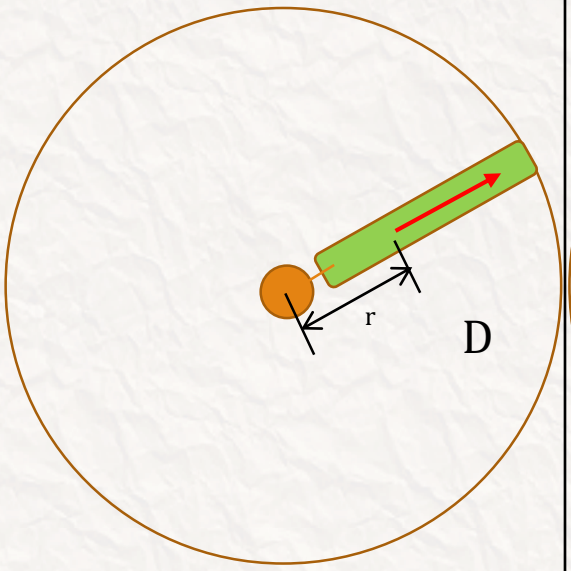
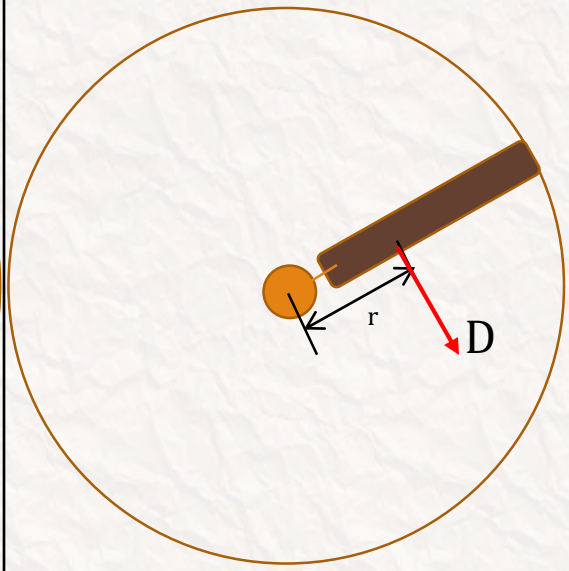
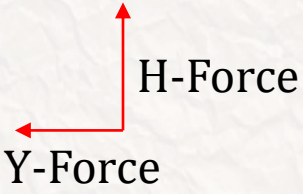
- Overall Rolling Moment:

$$\frac{b}{2\pi}\int_0^{2\pi}\left[\int_0^R L'r\sin\psi\,dr\right]d\psi$$

- Coning angle:

$$\beta_0 = \frac{1}{2\pi I\Omega^2}\int_0^{2\pi}\left[\int_0^R L'r\,dr\right]d\psi$$

In-Plane Forces

Blade Coning	Radial Skin Friction Forces	Drag
	 Generally Neglected	 

In-Plane Forces (Analytical Form)

- Deriving using the simplifying assumptions, we get:

$$C_{H_{TPP}} = \mu \frac{\sigma C_{d0}}{4} + \frac{\sigma a}{2} p \left[\frac{\mu \lambda_{TPP}}{2} \left(\theta_0 + \frac{\theta_{tw}}{2} \right) - \frac{\theta_{1c} \beta_0}{6} + \theta_{1s} \frac{\lambda_{TPP}}{4} + \frac{\mu \beta_0^2}{4} \right]$$

$$C_{Y_{TPP}} = \frac{\sigma a}{2} \left[\frac{3}{4} \mu \beta_0 \left(\theta_0 + \frac{2}{3} \theta_{tw} \right) + \frac{\theta_{1c}}{4} \lambda_{TPP} + \frac{\theta_{1s} \beta_0}{6} (1 + 3\mu^2) - \frac{3}{2} \mu \beta_0 \lambda_{TPP} \right]$$

Blade Flapping Dynamics (continued)

- Recall the blade flapping dynamics ODE

$$I\ddot{\beta} + I\Omega^2\beta = \int_{r=0}^{r=R} \frac{1}{2}\rho c(\Omega r + V_\infty \sin\psi)^2 r C_l dr$$

- and its solution of the form:

$$\beta = \beta_0 + \beta_{1c}\cos\psi + \beta_{1s}\sin\psi + \beta_{2c}\cos 2\psi + \beta_{2s}\sin 2\psi + \bullet\bullet\bullet$$

(Note that β_0 is the coning angle and $\psi = \Omega t$)

- Now, we also know the expression C_l

$$C_l = a\alpha_{effective} = a\frac{U_T\theta - U_P}{U_T} = \frac{a}{U_T}\begin{bmatrix} \Omega r\{\theta_0 + (\theta_{1c} - \beta_{1s})\cos\psi + (\theta_{1s} + \beta_{1c})\sin\psi\} \\ V\theta_0\sin\psi + V(\theta_{1c} - \beta_{1s})\cos\psi\sin\psi \\ + V(\theta_{1s} + \beta_{1c})\sin^2\psi - V\beta_0\cos\psi \\ - V\alpha_{TPP} - v \end{bmatrix}$$

Blade Flapping Dynamics (continued)

- Substituting, applying the assumptions, simplifying and separating terms, we get:

$$\beta_0 = \gamma\left[\frac{\theta_{80\%R}}{8}\left(1+\mu^2\right) - \frac{\mu^2}{60}\theta_{tw} - \frac{\lambda_{TPP}}{6} + \mu\frac{\beta_{1c}+\theta_{1s}}{6}\right]$$

$$\beta_{1c}+\theta_{1s} = \frac{-\frac{8}{3}\mu\left[\theta_{75\%R} - \frac{3}{4}\lambda_{TPP}\right]}{1+\frac{3}{2}\mu^2}$$

$$\beta_{1s}-\theta_{1c} = \frac{-\frac{4}{3}\mu\beta_0}{1+\frac{1}{2}\mu^2}$$

$$C_T = \frac{\sigma a}{2}\left[\frac{\theta_0}{3}\left\{1+\frac{3}{2}\mu^2\right\} + \frac{\theta_{tw}}{4}\left\{1+\mu^2\right\} + \mu\frac{\theta_{1s}+\beta_{1c}}{2} - \frac{\lambda_{TPP}}{2}\right]$$